

MUHAMAD AMIR ZULFIQAR BIN ZURAI

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EDUCATION

Bachelor of Computer Engineering with Honours (2022 - 2026)

Universiti Teknikal Malaysia Melaka (UTeM)

- Major in Computer Engineering

Foundation in Physical Science (2021-2022)

Universiti Teknologi Malaysia (UTM)

PROFESSIONAL EXPERIENCE

Internship as IT at Aplus Ventures

Jul 2025 - Oct 2025

- Developed a centralized Client Management System (CMS) using PHP and MySQL, consolidating records for 1000+ clients, business profiles, and licenses into a single platform.
- Enhanced operational workflow by implementing automated reminders and streamlining reporting, reducing manual data entry time by 40% and improving overall data accuracy.

WORK EXPERIENCE

Assistant Technician (Single-Phase Wiring – Part-Time)

9 Jun 2023 - 12 Jun 2023

- Partnered with senior technicians to execute the wiring and diagnostic maintenance of single-phase electrical systems across 2 commercial building sites.

Freelance Food Delivery Rider

May 2022 - Jan 2023

- Executed the secure and timely delivery of customer orders, averaging 15 to 20 completed trips per shift.

Catering Assistant

Oct 2021 - Dec 2021

- Executed the setup, service, and breakdown for 50+ corporate functions, weddings, and private parties, providing seamless service for events with up to 800 guests.

Bazaar Ramadan Food Assistant

Apr 2021 - May 2021

- Operated a fast-paced Bazaar Ramadan food stall, executing food preparation and fulfilling over 500+ customer orders per day.

ADDITIONAL INFORMATION

- **Languages:** Malay(native), English
- **Core Interests:** Scalable System Architecture, Advanced Data Processing, Embedded Systems and Hardware Integration.

AREA OF EXPERTISE

- **Programming & Web:**
 - Python, C/C++, Java, JavaScript, PHP, SQL, HTML, CSS.
- **Machine Learning & Data Science:**
 - TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, SHAP, LSTM, XGBoost, Time-Series Forecasting, RAG, Prompt Engineering.
- **Embedded Systems & IoT:**
 - FreeRTOS, STM32 Microcontrollers, Arduino, ARM, Multithreading & Semaphores, Hardware Interrupts (ISR), Blynk IoT.
- **Hardware & PCB Design:**
 - High-Speed PCB Design, Signal Integrity, Electromagnetic Interference (EMI), EasyEDA, Multisim, Transmission Line Theory.
- **Databases & Tools:**
 - MySQL, PostgreSQL, MongoDB, phpMyAdmin, Bootstrap, Git, Wireshark, Supabase, pgvector, and LangChain.

TECHNICAL PROJECTS

Ethereum Gas Price Forecasting Model (Final Year Project)

- Engineered a Hybrid XGBoost-LSTM machine learning architecture using TensorFlow and Scikit-learn to forecast Ethereum transaction fees 60 minutes in advance.
- Extracted and preprocessed 8,337 hourly on-chain data points via Etherscan and Binance APIs, engineering 30+ features including lag values and rolling statistics.

AI Customer Service Automator

- Engineered a production-ready Retrieval-Augmented Generation (RAG) Telegram bot using LangChain and Google Gemini 3.6 Flash to automate customer inquiries and menu management.
- Architected a vector database using Supabase (PostgreSQL + pgvector) to execute semantic search on inventory data.
- Deployed the application to the cloud via Render, integrating a lightweight Flask keep-alive server and automated pingging to achieve 24/7 uptime and eliminate cold-start latency.

Real-Time Petrol Station Simulation (RTOS)

- Architected a multi-pump petrol station simulation using three STM32F446RE Nucleo boards integrated with FreeRTOS.
- Implemented complex multithreading, hardware interrupts, and semaphores to manage 13 concurrent threads, ensuring synchronized, race-condition-free fuel level monitoring.

Tourist Arrival Forecasting Model

- Developed a multivariate Long Short-Term Memory (LSTM) neural network in Python to forecast international tourist arrivals across 6 major Malaysian states.

Solar-Powered Smart Irrigation Prototype

- Constructed an off-grid, solar-powered smart irrigation hardware system designed for sustainable agrivoltaic farming environments.
- Programmed Arduino microcontrollers and deployed the Blynk IoT application to process real-time soil moisture and environmental sensor data for automated water optimization.

EXTRA - CURRICULAR ACTIVITIES

Security Committee Member – Convocation Festival

Universiti Teknikal Malaysia Melaka (UTeM) | Nov 2025

- Coordinated with a 30-person security team to oversee campus traffic and crowd control throughout the 3-day convocation event.

Faculty Association Peer (Rakan Persatuan Fakulti)

Universiti Teknikal Malaysia Melaka (UTeM) | Oct 2024 – Feb 2025

- Partnered with a 30 persons committee to execute a series of 6 social and academic events, enhancing overall student engagement across the faculty.

Faculty Festival Committee Member

Universiti Teknikal Malaysia Melaka (UTeM) | 26 Oct 2024 – 27 Oct 2024

- Teamed with a 50-member committee to manage event logistics, directing crowd flow and vendor setup to maintain a strict event schedule.

Aerobics Instructor

Universiti Teknikal Malaysia Melaka (UTeM) | 22 Oct 2024

- Led high-energy group aerobics sessions for 50+ students and faculty members during a university-wide health and wellness event.

Futsal Team Manager – Sukan Antara Fakulti(SAF)

Universiti Teknikal Malaysia Melaka (UTeM) | 25 Nov 2024

- Managed end-to-end logistics for a 15-player futsal team, coordinating 6 training sessions and match schedules for the inter-faculty tournament.

Head of EXCO Multimedia

Universiti Teknologi Malaysia (UTM) | 2021-2022

- Directed the visual content strategy for 5 major university events, designing high-quality promotional materials that reached an audience of 100+ students.
- Coordinated a committee of 6 multimedia members, delegating 30+ design tasks and achieving a 100% on-time delivery rate for all promotional campaigns.